

Operating Systems: Three Easy Pieces

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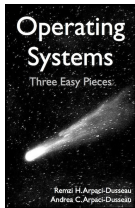
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Welcome to **Operating Systems: Three Easy Pieces** (now **version 1.10** -- see [book news](#) for details), a free online operating systems book! The book is centered around three conceptual pieces that are fundamental to operating systems: **virtualization**, **concurrency**, and **persistence**. In understanding the conceptual, you will also learn the practical, including how an operating system does things like schedule the CPU, manage memory, and store files persistently. Lots of fun stuff! Or [maybe not](#) so fun?

This book **is and will always be free** in PDF form, as seen below. For those of you wishing to **BUY** a copy, please consider the following:



- [Lulu Hardcover \(v1.10\)](#): this may be the best printed form of the book (it really looks pretty good), but it is also the most expensive way to obtain *the black book* of operating systems (a.k.a. *the comet book* or *the asteroid book* according to students). Now just: **\$39.75**
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Another way to help the book out: cite it! Here is the [BiBTeX entry \(seen below\)](#); you can also link to the site of the [best free operating systems book](#) on the market.

Operating Systems: Three Easy Pieces

Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau

Arpaci-Dusseau Books

November, 2023 (Version 1.10)

And now, the free online form of the book, in chapter-by-chapter form (now with chapter numbers!):

Intro

Virtualization

Concurrency

Persistence

Security

Preface	3 Dialogue	12 Dialogue	25 Dialogue	35 Dialogue	52 Dialogue
TOC	4 Processes	13 Address Spaces code	26 Concurrency and Threads code	36 I/O Devices	53 Intro Security
1 Dialogue	5 Process API code	14 Memory API	27 Thread API code	37 Hard Disk Drives	54 Authentication
2 Introduction code	6 Direct Execution	15 Address Translation	28 Locks code	38 Redundant Disk Arrays (RAID)	55 Access Control
	7 CPU Scheduling	16 Segmentation	29 Locked Data Structures	39 Files and Directories	56 Cryptography
	8 Multi-level Feedback	17 Free Space Management	30 Condition Variables code	40 File System Implementation	57 Distributed
	9 Lottery Scheduling code	18 Introduction to Paging	31 Semaphores code	41 Fast File System (FFS)	
	10 Multi-CPU Scheduling	19 Translation Lookaside Buffers	32 Concurrency Bugs	42 FSCK and Journaling	Appendices
	11 Summary	20 Advanced Page Tables	33 Event-based Concurrency	43 Log-structured File System (LFS)	Dialogue
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INSTRUCTORS: If you are using these free chapters, **please just link to them directly** (instead of making a copy locally); we make little improvements frequently and thus would like to provide the latest to whomever is using it. Also: we have made our own class-preparation notes available to those of you teaching from this book; please drop us a line at remzi@cs.wisc.edu if you are interested.

HOMEWORKS: Some of the chapters have homeworks at the end, which require simulators and other code. More details on that, including how to find said code, can be found here: [HOMEWORK](#)

PROJECTS: While the book should provide a good conceptual guide to key aspects of modern operating systems, no education is complete without projects. We are in the process of making the projects we use at the University of Wisconsin-Madison widely available; an initial link to project descriptions is available here: [PROJECTS](#). Coming soon: the automated testing framework that we use to grade projects.

BOOKS NEWS: Many small things to move to version 1.10. Track changes: [NEWS](#)

ACKNOWLEDGEMENTS: These students have greatly contributed to this effort, through endless bug reports and other comments. Your name could go here! (as well as in the printed book): [ERRATA](#)

OTHER SYSTEMS BOOKS: Interested in other systems books? Good! Of course, we assume some background in [The C Programming Language](#), so that's a good investment. And [Advanced Programming in the UNIX Environment](#) is a must for any shelf. On top of that, here are some OS books that could be worth your time: [Operating Systems: Principles and Practice](#) - [Operating System Concepts](#) - [Operating Systems: Internals and Design Principles \(8th Edition\)](#) - [Modern Operating Systems \(4th Edition\)](#) - [Linux Kernel](#)

[Development \(3rd Edition\)](#) - [Understanding the Linux Kernel](#) - [The Design and Implementation of the FreeBSD Operating System](#) - [Solaris Internals: Solaris 10 and OpenSolaris Kernel Architecture](#) - [Mac OS X Internals: A Systems Approach](#) - [The Design of the UNIX Operating System](#) - [UNIX: The Textbook](#) - [The Linux Programming Interface: A Linux and UNIX System Programming Handbook](#).

OTHER BOOKS: So you're looking down here? Well, how about reading something other than tech books all day long? Honestly, you need to be more balanced. Here are some awesome books you should most definitely read. Fiction: [Cloud Atlas: A Novel](#) - [Life of Pi](#) - [A Prayer for Owen Meany: A Novel](#) - [All the Light We Cannot See](#) - [The Book Thief](#) - [The Fault in Our Stars](#) - [Tenth of December: Stories](#) - [If I Don't Six](#) - [A Game of Thrones](#) - [To Kill a Mockingbird](#) - [The Kite Runner](#) - [Ender's Game](#) - [Foundation](#) - [Slaughterhouse-Five](#) - [The Shadow of the Wind](#) - [Flowers for Algernon](#) - [Holes](#) - [Atonement](#) - [The Name of the Wind](#) - [Beloved](#) - [For Whom the Bell Tolls](#) - [Different Seasons](#) - [Neuromancer](#) - [Snow Crash](#) - [Cryptonomicon](#) - [Shantaram](#) - [A Room with a View](#) - [Jude the Obscure](#) - [Illusions: The Adventures of a Reluctant Messiah](#) - [A Canticle for Leibowitz](#) - [A Wizard of Earthsea](#) - [Black Swan Green](#) - [The Stars My Destination](#) - [Ancillary Justice](#) - [My Brilliant Friend](#) - [Crossing to Safety](#) - [Possession](#) - [The Selected Works of T.S. Spivet](#) - [Essential Ellison](#) - [The Demolished Man](#) - [The Nightingale](#) - [The Overstory](#) - [The Windup Girl](#) - [The Water Knife](#) Non-fiction: [Seabiscuit: An American Legend](#) - [Unbroken](#) - [Surely You're Joking, Mr. Feynman!](#) - [On Intelligence](#) - [The Language Instinct](#) - [Flow](#) - [Guns, Germs, and Steel](#) - [The Selfish Gene](#) - [A Heartbreaking Work of Staggering Genius](#) - [Lies My Teacher Told Me](#) - [Freakonomics](#) - [How the Irish Saved Civilization](#) - [Cod](#) - [The Devil in the White City](#) - [The Swerve: How the World Became Modern](#) - [The Drunkard's Walk](#) - [The Visual Display of Quantitative Information](#) - [Eats, Shoots & Leaves](#) - [The Elements of Style](#) - [The Design of Everyday Things](#) - [Mountains Beyond Mountains](#) - [The Soul of A New Machine](#) - [Alan Turing: The Enigma](#) - [Consider the Lobster](#) - [The Vintage Guide to Classical Music](#)

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